Slide #1

Hi everyone, thank you for the warm introduction and for this opportunity to speak here today. I will be taking us on an exciting journey to the edge of the observable Universe, a journey in which we can explore the most distant galaxies using the James Webb Space Telescope, or JWST.

Slide #2

To set the stage, let me begin by answering an important question: What are galaxies? And why is it crucial to study the most distant galaxies?

Slide #3

Galaxies are immense cosmic islands. They contain lots and lots of stars, nebular gas, dust, and dark matter, all held together by gravity. Our own Milky Way, for example, is home to a few hundred billion stars.

Slide #4

One of those stars is our Sun, which is located roughly twenty-six thousand light-years from the Galactic Center in one of the minor spiral arms. For years, some astronomers argued that the Milky Way encompassed the entire Universe.

Slide #5

But roughly one hundred years ago, Edwin Hubble fundamentally transformed our understanding of the Universe by discovering that many of the mysterious "spiral nebulae" were incredibly distant, far beyond the spatial extent of the Milky Way, implying they were external galaxies. One of these external galaxies is our nearest major galactic neighbor, Andromeda. The Milky Way, together with Andromeda and a bunch of dwarf galaxies, form what we call the Local Group.

Slide #6

Soon after Hubble's initial discovery, astronomers searched for more galaxies and found that they were not uniformly distributed. Instead, these galaxies were clustered within massive gravitationally bound structures called galaxy clusters, which organized themselves into even larger structures, called superclusters.

Slide #7

This brings us from cartoon illustration to reality. Here is an actual map of the local, observable Universe using galaxies from the Sloan Digital Sky Survey, or SDSS. What you are seeing here is the so-called "cosmic web," which contains dense regions of space called "nodes," linear regions of space that connect the

nodes called “filaments,” and empty regions of space that separate the filaments called “voids.” Positioned at the nodes of the cosmic web are clusters and superclusters. Not only did SDSS reveal the large-scale distribution of galaxies it also revealed the incredible diversity of galaxies themselves, as shown by these images. Local galaxies have a wide range of luminosities, colors, morphologies, chemical compositions, star-formation rates, and masses.

Slide #8

Here is an even more impressive view of the observable Universe using 13 million galaxies from the Dark Energy Spectroscopic Instrument, or DESI. These galaxies span roughly 12 billion years of cosmic history, which corresponds to redshifts between 0 and 4.

Slide #9

What do I mean when I mention redshift? As light travels from great distances, it is stretched to longer and redder wavelengths by the Universe’s expansion. The greater the redshift, the greater the distance the light has traveled to reach us. Since space and time are interlinked, distant objects with increasing redshift are further back in time because it takes their light so long to reach us.

Slide #10

~~Why do we care about redshift and distant galaxies?~~ Astronomers believe that distant galaxies are fundamentally different from the modern galaxies observed locally. Most importantly, distant galaxies provide powerful constraints on our theories for galaxy formation and evolution, because these galaxies had less time to form and evolve when compared to the Milky Way and other local galaxies.

Slide #11

To quickly summarize the observed differences between local and distant galaxies, at fixed stellar mass, galaxies in the distant Universe typically have smaller comoving number densities, smaller physical sizes, smaller metallicities, larger specific star-formation rates, and stronger emission line equivalent widths.

Slide #12

Well, what about the first galaxies? This cartoon timeline shows that the Universe began nearly 13.8 billion years ago with the Big Bang and has been expanding ever since. After an initial period called the cosmic “dark ages,” the first stars and galaxies formed roughly 200 million years after the Big Bang, corresponding to a redshift of 20. These early sources started ionizing the islands of neutral hydrogen around them, triggering the Universe's last major phase transition –

reionization. When I say “ionize,” I mean that the neutral hydrogen gas filling the Universe was broken down into its constituent parts – protons and free electrons.

When I refer to the “distant” Universe, or, “Cosmic Dawn,” I am typically referring to this Epoch of Reionization at redshifts beyond 5 and 6, which has remained one of the least understood periods in cosmic history, primarily due to observational challenges. Overcoming these challenges to study galaxies in the first billion after the Big Bang required a revolutionary observational breakthrough.

Slide #13

This brings me to the second chapter of my talk where I will continue by answering another important question: What is the James Webb Space Telescope, or JWST? And how has it changed our view of the Universe?

Slide #14

JWST, or Webb, is a space telescope orbiting the Sun from a special location in space near the second Sun-Earth Lagrange point, called L2. Here is how JWST compares with the Hubble Space Telescope, or HST. The primary mirror is the most important component of a telescope for gathering light. The larger the mirror, the more light you can collect, and more distant objects can be detected. The primary mirror of JWST has an area 5 times greater than HST. It primarily operates at near- and mid-infrared wavelengths. At these wavelengths, JWST is so sensitive it can detect the heat from a bumblebee on the moon!

Slide #15

By viewing the Universe at infrared wavelengths with unprecedented sensitivity, JWST will open up a new window to the cosmos by directly observing some of the first stars and galaxies that formed after the Big Bang. Its infrared vision also allows JWST to study stars and planetary systems forming inside thick clouds of gas and dust that are opaque to visible light.

Slide #16

JWST has a suite of four powerful instruments that will investigate the cosmos. These four instruments include the Near Infrared Spectrograph (or NIRSpec), the Mid-Infrared Instrument (or MIRI), the Near Infrared Camera (or NIRCam), and the Near Infrared Imager and Slitless Spectrograph (or NIRISS). I spend most of my time using MIRI to study the most distant galaxies. As the only instrument on board JWST operating at mid-infrared wavelengths, it needs to be cryogenically cooled to roughly 10 degrees above absolute zero to minimize thermal emission.

Slide #17

These instruments were designed to answer outstanding questions about the Universe and to make breakthrough discoveries in all fields of astronomy. The two questions that I care most about are on the left side of this image: What did the early Universe look like? When did the first stars and galaxies form? More generally, how did galaxies and supermassive black holes form and evolve throughout cosmic time?

Slide #18

To demonstrate the superior light gathering capabilities of JWST, here I show an animation comparing false-color images from HST and JWST of a massive galaxy cluster at redshift 0.6. HST observed this deep field for more than one day, while JWST observed for less than two hours. Despite more than an order of magnitude difference in exposure time, JWST is able to spatially resolve the triply imaged galaxy MACS0647-JD at redshift 10.17, whereas HST could not. JWST also reveals relatively bright, red and yellow galaxies that were previously invisible to HST.

Slide #19

Here is one more animation demonstrating JWST's power, this time compared with the Spitzer Space Telescope, which was retired in 2020 just before the launch of JWST. Spitzer operated primarily at mid-infrared wavelengths, similar to MIRI. At these wavelengths, JWST's improvement is even more dramatic.

Slide #20

Let's quickly summarize the most important things I just discussed about JWST. JWST position at L2 gives it an unobstructed view of the Universe. JWST's large mirror size provides unprecedented light gathering capabilities, even compared to HST and Spitzer. JWST operates primarily at infrared wavelengths, which allows us to view the most distant objects.

Slide #21

This brings me to the third and final chapter of my talk where I will conclude by answering another important question: How is JWST being used to study the most distant galaxies? And what new questions have been raised?

Slide #22

I want to start by showing this figure from Dan Stark's recent review article, which illustrates the emission line redshift record over time in the left panel and the distribution of spectroscopic redshifts in the right panel, both before and after JWST's launch. In its first few years of operation, JWST has found hundreds

of galaxies beyond redshift 7 and dozens beyond redshift 10, while pushing the spectroscopic redshift record above redshift 14.

Slide #23

The most distant galaxy that we currently know about is MoM-z14, which emitted its light only 280 million years after the Big Bang, when the Universe was only 2% of its current age. The images shown here were produced using JWST's NIRCam instrument and were provided in a recent NASA press release. ~~This galaxy was found in the COSMOS legacy field and is located near the celestial equator.~~

Slide #24

The discovery of luminous galaxies like MoM-z14 with JWST was surprising and has challenged our understanding of galaxy formation and evolution in the early Universe since these luminous galaxies were observed to be more abundant than expected. To illustrate this overabundance of luminous galaxies, I am showing the ultraviolet (UV) luminosity function as measured at redshift 14 with JWST. The y-axis shows the number of galaxies per comoving volume while the x-axis shows the UV brightness of these galaxies. The blue point shows the observed value. The grey line demonstrates pre-JWST predictions for the UV luminosity function, which underpredicts the number of luminous galaxies at redshift 14 by more than two orders of magnitude. To put this another way, for every individual galaxy that we predicted to see, we are finding hundreds! The remainder of the lines show post-JWST predictions that implement modifications to the physics of early galaxy formation and evolution.

Slide #25

Why is this happening? We have a few different theories as shown in the legend on the preceding slide. I am focusing on three here...

1. Distant galaxies are predicted to experience intense episodic bursts of star formation. During these bursts, the youngest and most massive stars will dominate the light and "outshine" the older stellar population that dominates the mass. Galaxies will appear more luminous during bursts of star formation, while obscuring evidence of prior star formation.
2. Distant galaxies may have been able to convert nebular gas into stars much more efficiently. This increased star formation efficiency is possibly caused by feedback-free starbursts (FFBs), when the high densities and low metallicities at high redshifts guarantee increased SFEs in the most massive dark-matter halos due to ineffective feedback processes.
3. Distant galaxies are expected to form stars differently than galaxies in the local Universe, with IMFs that suppress low-mass star formation and enhance high-mass star formation due to lower metallicities and higher

temperatures of the cosmic microwave background (CMB), resulting in brighter galaxies and accelerated metal enrichment at fixed halo mass.

Slide #26

These three different theories have distinct predictions for the rest-frame optical properties of the most distant galaxies. However, beyond redshift 10, the highly diagnostic rest-frame optical wavelengths have redshifted beyond the spectral coverage of NIRCam and NIRSpec, and are only observable with MIRI. This is why my work focuses on using MIRI to study the most distant galaxies, to access these rest-frame optical wavelengths.

Slide #27

To quickly summarize what I just discussed: Here is a cartoon illustration of what we predicted to see prior to the launch of JWST...

Slide #28

And here is what we actually saw... Just to be clear, this cartoon is actually not as extreme as the discrepancy I showed a few slides ago.

Slide #29

Fundamentally, my research tries to answer the question of: "Why?" Why are there so many luminous galaxies in the early Universe? I will focus on answering this question for the remainder of my time today.

Slide #30

So what am I doing to answer this question? I am using observations from one of JWST's largest and most ambitious observing programs. This program is known as the JWST Advanced Deep Extragalactic Survey, or JADES, which is a large, international research group containing hundreds of astronomers. Using roughly 770 hours of observations from the instrument science teams, we point JWST at tiny, relatively empty regions of the night sky to produce "deep field images."

Many of the JADES folk are at my former institution, the University of Arizona, including numerous members from JWST's instrument teams. These people were responsible for creating and calibrating some of the instruments on board JWST. From these experts, I learned how to push the limitations of NIRCam and MIRI.

Slide #31

Within a few months of science operations, in December of 2022, the JADES team spectroscopically confirmed the three most distant galaxies known at the time, thereby reaching a new milestone in the redshift frontier.

On the left, I am showing the NIRCam imaging used to select these sources as high-redshift galaxy candidates. This is the same image I was showing on the previous slide. To give a sense of scale, this image is the same size on the night sky as a human would appear from a mile away.

On the right, I am showing the NIRSpec spectroscopy used to confirm the distances to these high-redshift galaxies. We measured distances using a feature called the Lyman break. This feature is caused by absorption of UV photons from neutral hydrogen in the intergalactic medium. The redshifts of these four galaxies decrease from top to bottom.

Slide #32

This figure shows a zoom-in of the NIRCam images shown in the previous slide. These images were taken in the GOODS-South field; GOODS stands for the Great Observatories Origins Deep Survey, an extragalactic deep field encompassing the Hubble Ultra Deep Field, or HUDF. These fields are near the Constellation Fornax.

Slide #33

I want to take a moment to zoom in, pause, and appreciate the beauty of this image, an image that contains more than 45,000 galaxies. Nearly all of these galaxies emitted their light billions of years ago! Many of these galaxies were invisible prior to the launch of JWST...

Slide #34

Let's take a look at some of those galaxies that were identified and confirmed by JADES. All of the images I am about to show are publicly available on the JADES website using the link provided here. This is the first galaxy ever confirmed to be beyond redshift 12, JADES-GS-z12-0 at z = 12.48, which emitted its light roughly 345 million years after the Big Bang. This galaxy is small and faint: 29th magnitude, containing roughly 100 million solar masses in stars within a diameter of roughly 660 light years. NIRSpec data revealed an emission line from doubly ionized carbon in the rest-frame UV for this galaxy.

Slide #35

Here is the first galaxy ever confirmed to be beyond redshift 13, JADES-GS-z13-0 at z = 13.16, which emitted its light 320 million years after the Big Bang. This was

the record holder and most distant galaxy that we knew about for more than a year. Once again, this galaxy is small and faint: less than 29th magnitude, containing roughly 60 million solar masses in stars within a diameter of roughly 400 light years. NIRSpec data did not reveal any emission lines for this galaxy.

Slide #36

Roughly a year and a half ago, in August of 2024, we spectroscopically confirmed another distant galaxy: JADES-GS-z13-1 at z = 13.05, emitting light 325 million years after the Big Bang. Once again, this galaxy is small and faint: less than 29th magnitude, containing roughly 50 million solar masses in stars within a diameter of roughly 320 light years. Surprisingly, we detected the hydrogen recombination line Lyman-α in this galaxy, which is the most distant detection of this emission feature; this tells us something new about the patchiness of reionization. Please ask me about this galaxy after my talk if you are interested in learning more.

Slide #37

Just a few months before that, in May of 2024, we spectroscopically confirmed the next farthest galaxy: JADES-GS-z14-1 at z = 13.90, emitting light 300 million years after the Big Bang. Once again, this galaxy is small and faint: less than 29th magnitude, containing roughly 40 million solar masses in stars within a diameter of roughly 320 light years. So far, all of the galaxies that I have shown you are the types of galaxies that we expected to observe with JWST: small, faint, spatially unresolved when observed by NIRCam, and undetected when observed by MIRI.

Slide #38

However, at the same time as discovering the previous galaxy, we found the unexpected when we spectroscopically confirmed the first galaxy beyond redshift 14: JADES-GS-z14-0 at z = 14.18, emitting light 290 million years after the Big Bang. This was the record holder and most distant galaxy that we knew about for roughly a year. Unlike those last four galaxies, this galaxy is neither small nor faint: 27th magnitude, containing roughly 600 million solar masses in stars within a diameter of roughly 1,700 light years. You will notice that this is not just one galaxy, but two galaxies: a low-redshift Balmer-break galaxy at z = 3.47 and a high-redshift Lyman-break galaxy at z = 14.18. On one hand, we got incredibly lucky to find such a bright galaxy within the small area of JADES. On the other hand, we were incredibly unlucky to have such a distant galaxy right beside a foreground galaxy. And then we were even more unlucky because a Balmer-break at z = 3.47 falls within the same NIRCam filter as a Lyman-break at z = 14.18. Because of this, some of us joke by saying that the Universe was playing a cruel trick on us. But despite the Universe's cruelty, this is a very popular galaxy: there are already a few dozen papers written exclusively about JADES-GS-z14-0.

Slide #39

I wanted to show this screenshot of a Slack message that one of my PhD advisors, Kevin Hainline, sent me on February 7th, 2023. This is the first ever message about JADES-GS-z14-0! We were the very first people to lay eyes on this galaxy. At the bottom, I am showing the NIRCam images used to select this source. As shown at the top, Kevin estimated the redshift by fitting these images with the computer software EAZY. I would agree with the contents of this message: JADES-GS-z14-0 is indeed a true weirdo!

Slide #40

Scientific American also agrees that this galaxy is a true weirdo. Our discovery received lots of international press, including a NASA blog post.

Slide #41

Let's get back to science results. Here is the ten hour spectrum that we originally used to confirm the redshift and distance to this galaxy using the Lyman break. As you can see, the spectrum is essentially featureless because we are looking at rest-frame UV wavelengths. As I mentioned before, the rest-frame optical lines that we know and love have been redshifted into the wavelength coverage of MIRI, beyond the wavelength coverage of NIRCam and NIRSpec, for this galaxy.

Slide #42

Thankfully, JADES has an ultra-deep MIRI imaging program at 7.7 microns in GOODS-S with total on-source integration times of 43 hours. The same region of the sky, known as the JADES Origins Field or JOF, also has an ultra-deep NIRCam imaging program with total on-source integration times of 167 hours split between 14 filters. These two programs total nearly 9 days of observing time in a tiny region of the sky, covering only 10 square arcminutes, just two NIRCam modules. The RGB illustrates the MIRI footprint and combines the NIRCam and MIRI observations. We got incredibly lucky: JADES-GS-z14-0 falls within both of the ultra-deep imaging programs, right at the edge of the MIRI footprint.

Slide #43

Let's take a closer look at some of these images. The blue image is one of the short-wavelength NIRCam filters at 1.15 microns, where JADES-GS-z14-0 is invisible. The green image is one of the long-wavelength NIRCam filters at 2.77 microns, where this galaxy is visible and bright. The red image is the MIRI filter at 7.7 microns, where it is ridiculously bright, brighter than all of the NIRCam filters. The RGB from these three filters is shown in the bottom right. If you remember those other distant galaxies I showed you earlier, none of them were detected, or visible, with MIRI. But not JADES-GS-z14-0!

Slide #44

Let's look at the spectral energy distribution, which shows the photometric measurements in blue from all of the available imaging filters. In the upper panel, on the y-axis, we have the flux density in nano-Janskies. On the x-axis, we have the observed wavelength in microns. We performed complicated modeling of the data by combining stars, dust, and nebular gas. Our best-fit models are shown in gray. In the lower panels, we provide the residuals between the best-fit models and the data, along with the filter transmission curves. As mentioned before, rest-frame optical wavelengths are redshifted into the spectral coverage of MIRI. The 7.7 microns data point contains bright emission lines from recombining hydrogen and doubly ionized oxygen: Hβ and [OIII5007. Roughly one-third of the MIRI flux actually comes from these emission lines. This is ridiculous: nebular gas, consisting of hydrogen and oxygen, which has been detected with MIRI in a galaxy beyond redshift 14. This surpassed our wildest dreams for MIRI!

Slide #45

Most importantly, we were able to indirectly provide constraints on the oxygen abundance using the MIRI photometric detection. JADES-GS-z14-0 is metal enriched. The oxygen lines are stronger than the hydrogen line; the ratio between these lines is shown on the x-axis, which is often used as a proxy for metallicity. JADES-GS-z14-0 has smaller oxygen abundance than typical values for galaxies at redshift 8, when the Universe was roughly twice as old. The main takeaway is that our results suggest rapid mass assembly and metal enrichment during the earliest phases of galaxy formation.

Slide #46

However, when we modeled the observations shown in grey here, we found another solution when using another piece of code, as shown by the red curves. This alternative solution attributed the excess 7.7 microns flux to a strong continuum break with weak emission lines rather than a weak continuum break with strong emission lines, as shown by the blue curves. If the red model was true, JADES-GS-z14-0 would be almost an order of magnitude more massive than suggested by the blue model and contain an extremely evolved stellar population. This would have huge implications for our understanding of the early Universe and might even require modifications to our standard concordance cosmology.

Slide #47

Despite nearly 10 days of JWST mission time spent observing JADES-GS-z14-0, the most important question remained: What was the source of the excess 7.7 microns flux? Strong emission lines? Or a strong continuum break?

Slide #48

We received a preliminary answer to this question using follow-up observations at millimeter wavelengths from ALMA. These observations detected an emission line from doubly ionized oxygen at rest-frame 88 microns.

Slide #49

However, to directly answer the question regarding the source of the excess 7.7 microns flux, we desperately needed spectroscopic observations at mid-infrared wavelengths. I submitted a successful JWST Cycle 4 proposal to obtain these observations, which were acquired a few months ago in November of 2025. At roughly 63 hours of mission time for an individual spectrum of an individual galaxy, this is among the most audacious and absurd programs ever approved.

Slide #50

When these observations were acquired, I ran into a ton of issues with data processing, as summarized on this slide. I do not expect you to read all of these words. The primary thing that I want you to take away is: JWST/MIRI is arguably the least understood instrument on board JWST; the MIRI/LRS spectrum that I obtained is the longest and most technically challenging observing program ever conducted by MIRI.

Slide #51

To give you a sense of the data I am working with, here are the raw and partially processed files obtained by JWST. I needed to convert hundreds of gigabytes of these images into a single spectrum of less than a few hundred kilobytes.

Slide #52

To complicate things even further, during normal solar activity, at least half of the pixels in each image are affected by cosmic ray events. However, when my observations were acquired, the Sun was not behaving normally. Does anyone remember the Northern Lights that were visible here in November of last year? I definitely remember this, because they affected my observations with orders of magnitude more cosmic rays than expected!

Slide #53

Most notably, the elevated solar activity ruined one-third of my observations; nearly one full day of mission time wasted. This was caused by a telescope failure where JWST performed target acquisition, or TA, on a cosmic ray rather than the star that I picked out, which is at the center of these images.

Slide #54

Thankfully, two-thirds of the observations were successful, as demonstrated by this figure showing the on-sky locations of the slits that we used. The galaxy that we care about, JADES-GS-z14-0, is indicated by the white cross. The failed set of observations are shown at the bottom, which completely missed our galaxy.

Slide #55

Before I do the big reveal of the spectrum, I want to remind everyone of the question we are trying to answer: What was the source of the excess 7.7 microns flux for JADES-GS-z14-0? Strong emission lines? Or a strong continuum break?

Slide #56

And here it is! The upper panel shows the two-dimensional spectrum. We used two nods for our observations, as demonstrated by the dark traces. We combined these two nods to produce the central, bright trace. The blue horizontal lines roughly indicate the extraction profile used to produce the one-dimensional spectrum shown in the bottom panel. The blue vertical lines indicate the locations for some of the brightest rest-frame optical emission lines.

There are a few things I want to point out about this spectrum in the bottom panel. First and foremost, we have a strong detection of the [OIII] doublet around 7.5 microns, confirming that the 7.7 microns flux excess is from strong emission lines. These oxygen lines are even stronger than we expected! Furthermore, we have a strong detection of Halpha around 10 microns. Finally, we have a strong detection of the rest-frame optical continuum between 5 and 7 microns, with tentative evidence for a Balmer continuum break. If confirmed, a Balmer break would indicate a significant population of old stars within JADES-GS-z14-0.

Slide #57

So what did we learn from these observations? This figure shows a comparison with other galaxies beyond redshift 3 observed by JWST. On the y-axis, I am showing the Halpha line luminosity while on the x-axis, I am showing the UV continuum luminosity. The green and grey points represent galaxies beyond redshift 3 from JADES. The purple diamonds represent three luminous galaxies beyond redshift 10 that have MIRI spectroscopy. Finally, the blue star represents JADES-GS-z14-0.

Halpha is a hydrogen recombination line that is often used as a proxy for the recent star-formation rate, as indicated by the twin y-axis on the right side of the figure. JADES-GS-z14-0 is forming roughly ten solar masses per year in stars,

which is smaller than previously inferred from the SED fitting; this is telling us something about the burstiness of the star-formation history for this galaxy.

Related to the burstiness of the star-formation history, the brown diagonal lines show different values of the ionizing photon production efficiency, or xi_ion, which is found by taking the ratio of the Halpha and UV luminosities. This is telling us about the number of ionizing photons produced per unit luminosity.

Slide #58

Here is a similar figure as the one shown on the previous slide, but now with the ionizing photon production efficiency shown on the y-axis. The value that we measure for JADES-GS-z14-0 is similar to the canonical value that we observe locally, which is shown by the grey shaded region. It is somewhat lower than typical values measured for galaxies beyond redshift 3 and much lower than the stellar population maximum, which is shown by the black line.

Slide #59

This figure shows different emission line ratios on the x- and y-axes. The O32 index is on the y-axis and is often used as a proxy for the ionization parameter, while the R3 and R23 indices on the x-axes are often used as a proxy for the gas-phase metallicity. These emission line ratios indicate extreme ionization conditions and metallicities consistent with roughly 10% solar.

Slide #60

However, when you use the emission line ratios shown on the previous slide to infer physical properties, you are only using two or three of the emission lines that we have observations for. We can instead use detailed photoionization modeling to infer physical properties, as demonstrated by this figure. If you fit all of the emission lines shown in the upper right panel, we instead infer metallicities consistent with roughly 50% solar, and even up to 100% solar, less than 300 million years after the Big Bang.

Slide #61

To put it another way, we have two different measurements for the chemical abundance pattern and metallicity of JADES-GS-z14-0. The one using standard strong-line diagnostics is consistent with the predicted yields from core collapse supernova, as shown by the grey shaded region, while the other one using detailed photoionization modeling is more consistent with galactic chemical evolution and values measured locally, as shown by the black curve.

Slide #62

The metallicities inferred for JADES-GS-z14-0 are surprising and astonishing so early in cosmic history. Why are they surprising? Because such high metal production requires multiple generations of massive stars to have lived, died, and exploded, thereby enriching the surrounding interstellar material. This all needs to happen within a few hundred million years of the Big Bang.

Slide #63

So, is it possible to have such extreme metallicites? This figure demonstrates predictions from chemical evolution models assuming typical values for chemical yields and three different star-formation histories. To reproduce the metallicity inferred from standard strong-line diagnostics, as shown in the upper panels, we need to assume somewhat large star-formation efficiencies. But to reproduce the metallicity inferred from detailed photoionization modeling, as shown in the lower panels, we need to assume much higher star-formation efficiencies.

Slide #64

With this in mind, let me return to the question I posed earlier, the fundamental question driving my research about understanding galaxies in the early Universe: Why have we discovered an overabundance of luminous galaxies beyond redshift 10? My ongoing work with JADES-GS-z14-0 suggests the answer is both bursty star-formation histories and increased star formation efficiencies.

Slide #65

Bursty star-formation histories can also easily explain the puzzle of why there are two distinct classes of luminous galaxies beyond redshift 10. These two classes of objects include extended galaxies with relatively weak emission lines as shown in orange, such as JADES-GS-z14-0, and compact galaxies with relatively strong emission lines as shown in blue, such as MoM-z14.

Slide #66

So how does bursty star-formation histories explain the differences in spatial extent? Outshining. The idea is compact galaxies like MoM-z14 are experiencing a central starburst that decreases the observed size of the galaxy. Eventually, inside-out quenching occurs when feedback becomes effective, which eventually produces a spatially extended and mini-quenched galaxy, like JADES-GS-z14-0.

Slide #67

And how does bursty star-formation histories explain the differences in emission line strengths at fixed UV continuum luminosity? The idea is galaxies with strong

emission lines have experienced a burst of star-formation in the last few million years, while galaxies with weak emission lines have experienced a recent lull or quench in the last few million years.

Slide #68

This brings me to the end and final few slides of my talk. I want to briefly turn our attention to the future of studying the most distant galaxies...

Slide #69

In the long term, I am excited for the upcoming Roman Space Telescope, which will address some of JWST's limitations by having an expansive field-of-view and rapid surveying capabilities, albeit with worse sensitivity, spectral coverage, and spatial resolution when compared to JWST. Roman is expected to observe tens of thousands of galaxies beyond redshift 10. These samples will be ideal targets for follow-up observations at rest-frame optical wavelengths with JWST.

Slide #70

I will conclude by leaving you all with the brief summary and central thesis of my talk today. The most distant galaxy candidates are found with JWST's NIRCam instrument, and eventually will be found with Roman. Then they are confirmed with NIRSpec or ALMA. But they are not fully understood without JWST/MIRI. MIRI is still the future for understanding the first stars and galaxies!

Slide #71

As we watch this fly-through visualization to JADES-GS-z14-0, I am happy to take any questions you might have. Please ask me about preliminary ongoing work about JADES-GS-z14-0, if you are interested. Thank you so much for your time.

Slide #72

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